The 69th Annual Meeting of the AustMS – overview

For more detailed information, refer to the conference website: austms2025.org.au.

Plenary speakers

Abstracts are available at austms2025.org.au/plenary-speakers/

- **John Bamberg** (University of Western Australia)
- **Agnese Barbensi** (University of Queensland)
- **Tobin Driscoll** (University of Delaware, USA)
- **Jared Field** (University of Technology Sydney)
  Dr. Yunupingu lecturer
- **Cecilia Gonzalez-Tokman** (University of Queensland)
  ANZIAM lecturer
- **Sophie Hautphenne** (University of Melbourne)
- **Theo Johnson-Freyd** (Dalhousie University / Perimeter Institute, Canada)
  ANZAMP lecturer
- **Anita Liebenau** (UNSW Sydney)
  Hanna Neumann lecturer
- **Yoshihiro Tonegawa** (Institute of Science Tokyo, Japan)
- **Caroline Wormell** (University of Sydney)
  Early career lecturer
- **Oksana Yakimova** (University of Jena, Germany)

Special sessions

- Algebra and Combinatorics
- Applied and Industrial Mathematics
- Category Theory, Homotopy Theory, and K-Theory
- Computational Mathematics
- Differential Geometry and Symmetry
- Dynamical Systems and Ergodic Theory
- Equity, Diversity and Inclusion
- Financial Mathematics
- General Session
- Groups, Actions and Computation
- Harmonic Analysis and Hamiltonian PDE
- Mathematical Methods in Continuum Mechanics and Wave Theory
- Mathematical Physics
- Mathematics Education
- Nonlinear PDEs and their applications
- Number Theory
- Optimisation
- Probability and Mathematical Statistics
- Representation Theory
- Stochastic Differential Equations
- Topology
- Visualisation, Mathematical Art, Engagement and Outreach

Schedule

Monday December 8

4:00pm	WIMSIG Business Meeting	Borchardt Library, Room 1.34
6:00pm	WIMSIG Dinner	Glenn College Dining Hall

Day 1 — Tuesday December 9

8:00am	Information and Registration desk opens	Agora East, Level 2
9:00am	Opening ceremony and prizes	Agora Theatre
11:00am	Morning tea	Agora East, Level 2
11:30am	**Plenary Talk – Tobin Driscoll** *Computational approximation by rational functions*	Agora Theatre
12:30pm	Lunch break	
1:30pm	Special sessions	Various locations
3:30pm	Poster session	Agora East, Level 2
3:30pm	Afternoon tea	TLC, Level 2 and Agora East, Level 2
4:00pm	Special sessions	Various locations
5:30pm	**Plenary Talk – Sophie Hautphenne** *Branching processes in practice: models, inference, and applications*	Agora Theatre
6:30pm	Welcome Reception	Jenny Graves Lawn

Day 2 — Wednesday December 10

8:30am	Information and Registration desk opens	Agora East, Level 2
10:00am	**Plenary Talk – Agnese Barbensi** *Mapping the topology of polymer folding: knots, geometry, and data*	Agora Theatre
11:00am	Morning tea	Agora East, Level 2
11:30am	**Plenary Talk – Yoshihiro Tonegawa** *Geometric measure theory and mean curvature flow*	Agora Theatre
12:30pm	Lunch break	
1:00pm	Debate: AI will destroy mathematics	Agora Theatre
2:00pm	Special sessions & Education Afternoon	Various locations
3:30pm	Poster session	Agora East, Level 2
3:30pm	Afternoon tea	TLC, Level 2 and Agora East, Level 2
4:00pm	Special sessions & Education Afternoon	Various locations
5:30pm	Refreshments Before the Public Lecture	Agora East, Level 2
6:00pm	**Public Lecture – Tobin Driscoll** *Why Your Calculator Is (Slightly) Lying to You—And Why That's OK*	Agora Theatre

Day 3 — Thursday December 11

8:30am	Information and Registration desk opens	Agora East, Level 2
9:00am	**Plenary Talk – Cecilia Gonzalez-Tokman** *A journey through random dynamical systems and multiplicative ergodic theory*	Agora Theatre
10:00am	Morning tea	Agora East, Level 2
10:30am	**Plenary Talk – John Bamberg** *Finite generalised polygons: the final frontier in the classification of spherical buildings*	Agora Theatre
11:30am	**Plenary Talk – Oksana Yakimova** *Commutative subalgebras of the enveloping and symmetric algebra of a Lie algebra* $\mathfrak{q}$	Agora Theatre
12:30pm	Lunch break LGBTQIA+ and Allies Lunch	Agora East, Level 2
1:30pm	Special sessions	Various locations
3:30pm	Afternoon tea	TLC, Level 2 and Agora East, Level 2
4:00pm	AustMS Annual General Meeting	Agora Theatre
7:00pm	Conference dinner	Leonda by the Yarra

Day 4 — Friday December 12

8:30am	Information and Registration desk opens	Agora East, Level 2
9:00am	**Plenary Talk – Caroline Wormell** *Computing transfer operators and fractal dimensions*	Agora Theatre
10:00am	Morning tea	Agora East, Level 2
10:30am	**Plenary Talk – Anita Liebenau** *From Graphs to Latin Rectangles: Probabilistic Methods in Asymptotic Enumeration*	Agora Theatre
11:30am	**Plenary Talk – Theo Johnson-Freyd** *Fermions and categorified algebra closure*	Agora Theatre
12:30pm	Lunch break	
1:30pm	Special sessions	Various locations
3:30pm	Afternoon tea	TLC, Level 2 and Agora East, Level 2
4:00pm	Special sessions	Various locations
6:00pm	Conference concludes	